PHY · Gravity, projectiles and orbits — syllabus points and past-paper evidence

Every award of the 2016–2025 papers, labelled to the 2026/2027 Year 12 syllabus content (identical in both). Statistics from the labels; notes written by AI from that evidence: review before use.

Pattern 1: Newton's law of universal gravitation

Syllabus: S26-U3-SU-07 / S27-U3-SU-07

Frequency: 3 of 10 exam years; 15 marks in all.

Marks reward: Qualitative explanation 5, Select the principle or relationship 3, Graph work 3, Substitution and working 2, Recall 1, Final answer 1

Past questions: 2017P-16(bii); 2017P-17(a); 2017P-20(d); 2018P-12(c); 2018P-13(a); 2022R-8(a); 2022R-8(b)

Summary: This point covers the attraction between any two masses, F = Gm1m2/r^2, and how it is used in calculations, in explanations of when gravity can be ignored, and in sketching how the force falls off with separation. In the papers it usually appears as a short numerical calculation, a justification of negligible gravity, or a force-versus-separation graph, with a principle mark for linking gravity to orbital motion.

Recognise: Look for two masses (spheres, planets, subatomic particles, spacecraft) with a stated separation, a request to calculate or compare a gravitational force, a request to justify why gravity is negligible, or a request to show how force changes as objects move apart. Questions about orbital speed that set gravity equal to centripetal force also use this point.

Steps

1. Identify the two masses and the separation. For spheres, r is measured between centres, so convert a given diameter into a centre-to-centre distance where needed.
2. Convert all quantities to SI units: masses in kg and distance in m, watching for cm or mm.
3. Substitute into F = G m1 m2 / r^2 using G = 6.67 x 10^-11 N m^2 kg^-2, keeping the square on r.
4. Evaluate and state the answer in newtons with a sensible number of significant figures. Gravity is attractive and acts along the line joining the centres.
5. For a force-separation graph, use the inverse-square relationship: doubling r quarters F. Calculate several points, plot them, and join them with a smooth decreasing curve that approaches the axis but never touches it.
6. To justify ignoring gravity, state that the force exists, then compare its size with the precision needed or with other forces, supporting it with a formula and a calculated value.
7. For orbits, equate gravitational force to centripetal force m v^2 / r and rearrange to find the speed, e.g. v = sqrt(GM/r).

Where the marks are

- A mark for stating that gravitational attraction exists between any two masses, before using the formula.
- Marks for correct substitution of G, both masses and r^2 with units, and a correct final value with the right unit and significant figures.
- Marks for graph work: plotting the required points, labelling values on the force axis, and drawing a smooth inverse-square curve.
- A mark for explaining negligibility by quantifying the force against the precision required, not just saying it is small.
- A principle mark for setting gravitational force equal to centripetal force when deriving an orbital quantity.

Pitfalls

- Using the surface-to-surface gap of spheres instead of the centre-to-centre distance.
- Dropping the square on r, or mishandling the power of ten in G, which gives an answer off by many orders of magnitude.
- Drawing a straight line or a curve that rises or touches the axis when sketching force against separation.
- Saying gravity is small or negligible without a formula or a calculated comparison.
- Claiming gravity can be ignored in orbital problems where it is the force providing the centripetal acceleration.

Pattern 2: Gravitational fields around masses

Syllabus: S26-U3-SU-08 / S27-U3-SU-08

Frequency: 0 of 10 exam years; 0 marks in all.

Summary: This point asks students to describe the gravitational field that any mass produces in the space around it, including its direction, how its strength depends on distance, and how fields from several masses combine. No past-paper parts exist for this point (parts_total 0), so this note is written from the physics alone and the typical exam forms are inferred rather than observed.

Recognise: Look for wording such as 'gravitational field', 'field strength', 'g', 'describe the field around a planet', 'sketch the field lines', 'how does g change as the distance from the centre increases', or a question comparing the field at two distances or from two masses. Any question asking for a force per unit mass on a small test mass is also testing this idea.

Steps

1. Identify the mass producing the field (M) and the point of interest, and measure the distance r from the centre of that mass to the point, not from its surface unless the radius is given.
2. Use the field strength relation g = F/m = GM/r^2, where G is the universal gravitational constant. Field strength is in N kg^-1.
3. State the direction: the field at every point points radially towards the centre of the mass, because gravity is always attractive.
4. For a change in distance, compare by ratio: doubling r makes g one quarter of its previous value, and halving r makes it four times larger. Inverse square means the change is not linear.
5. For two or more masses, find the field from each mass separately, as vectors with direction, and add them vectorially at the point. Do not simply add magnitudes unless the fields are in the same direction.
6. For a sketch, draw radial arrows pointing into the mass, with lines closer together near the mass to show the stronger field, and spread further apart farther away.
7. Link to weight where relevant: the weight of a mass m at that point is W = mg, so a stronger field gives a larger weight.

Where the marks are

- A mark for stating that the field points towards the mass, not away from it.
- A mark for using the centre-to-centre distance in g = GM/r^2 and showing the inverse-square relationship.
- Marks for a correct sketch: radial arrows, correct direction, and spacing that shows the field weakens with distance.
- A mark for correctly comparing two field strengths using a ratio, rather than estimating the change.
- A mark for adding fields as vectors with direction when more than one mass contributes.

Pitfalls

- Measuring r from the surface of the object instead of from its centre, which gives a wrong value for g.
- Drawing field lines or arrows pointing away from the mass, confusing gravity with a repulsive effect.
- Assuming g is constant at all distances, when it actually falls off with the square of distance.

- Giving the field a unit of newtons instead of N kg^-1, or mixing up field strength with force.
- Adding magnitudes of field contributions from two masses without considering direction, which gives an incorrect resultant.

Pattern 3: Weight in a gravitational field (F = mg)

Syllabus: S26-U3-SU-09 / S27-U3-SU-09

Frequency: 6 of 10 exam years; 19 marks in all.

Marks reward: Substitution and working 7, Diagram 5, Select the principle or relationship 5, Final answer 1, Recall 1

Past questions: 2016P-15(a); 2016P-18(d); 2017P-20(c); 2019R-7(b); 2020C-21(di); 2020P-13(a); 2020P-13(b); 2020P-13(d); 2020R-11(a); 2020R-11(b); 2021P-14(ci); 2021R-9; 2023P-16(b); 2023R-12

Summary: This point tests the idea that weight is a gravitational force that exists because an object sits in a gravitational field, and that its size is F = mg. It rarely stands alone: it usually appears as a free body diagram, a force balance, or an earlier step inside a longer motion, ramp, or charged-particle problem.

Recognise: Look for any question that mentions weight, the mass of an object in kilograms, or asks for a free body diagram of an object near Earth. Also watch for questions where the weight must be found first before a tension, normal force, net force, acceleration, or electric force can be balanced against it.

Steps

1. Identify the object and write its mass in kilograms, converting from grams if needed.
2. Use weight = mass x gravitational field strength, F = mg, with g about 9.8 N kg-1 near Earth's surface, and state the result in newtons.
3. Draw the weight as a single arrow from the centre of the object pointing vertically downward toward the centre of the source of gravity.
4. If the object is on a slope or hung at an angle, resolve the other forces into vertical and horizontal components, and keep the weight vertical.
5. Write the balance or Newton's second law equation in the chosen direction, such as the vertical sum of forces equal to zero for equilibrium, or net force equal to ma when accelerating.
6. Substitute mg into that equation, solve for the unknown, and check that the answer has units of newtons or the required quantity.

Where the marks are

- A mark for drawing a single labelled weight arrow pointing straight down from the object, and for keeping equal-sized forces the same length when several objects are shown.
- A mark for stating the relationship F = mg with the mass in kg and g near 9.8 N kg-1, showing the substitution clearly.
- A mark for writing a vertical force equation that includes the weight term, such as the upward force equal to mg in equilibrium.
- A mark for a correct numerical answer with the correct unit, consistent with the substitution shown earlier.
- A mark for naming weight as the gravitational force that acts on an object in a gravitational field.

Pitfalls

- Treating mass in kilograms as if it were weight in newtons, or substituting m where mg is needed.
- Drawing the weight arrow in a direction other than straight down, or leaving it out of a free body diagram.
- Drawing the net force arrow when the question asks only for the individual forces, or adding extra forces that do not act on the object.
- Failing to resolve the weight into components when the object is on an incline, or resolving the wrong component.
- Dropping units or using grams directly in F = mg, which gives an answer that is wrong by a factor of

1000.

Pattern 4: Work and gravitational potential energy

Syllabus: S26-U3-SU-10 / S27-U3-SU-10

Frequency: 4 of 10 exam years; 10 marks in all.

Marks reward: Substitution and working 4, Final answer 3, Select the principle or relationship 2, Units, significant figures, direction 1

Past questions: 2016R-2; 2019P-12(d); 2021R-10(a); 2025P-15(a)

Summary: This point tests how the change in gravitational potential energy of a mass moving in a field is found, and how that energy links to work done and to kinetic energy. It appears mainly as a short calculation of energy gained or lost when a body is lifted or falls through a height, and in multi-part questions where energy conservation is used to find a speed or a derived expression.

Recognise: Look for a mass moving between two heights or positions near a planet, a body lifted or dropped, a climber or pipe-and-magnet style setup, or a question asking for gain or loss of potential energy, work done against gravity, power supplied, or a speed found from a height. Any question that gives mass and height change and asks for energy, or asks for a speed after a fall, is this point.

Steps

1. Identify the initial and final positions and find the change in height, delta h = h_final - h_initial, working out the sign from the direction of motion.
2. Use the near-surface relation for gravitational potential energy change: delta Ep = m g delta h, with g = 9.8 m s-2 or as given in the question.
3. If the question asks for work, use W = F s, or equivalently W = delta E, and recognise that the energy transferred equals the change in stored energy.
4. For a speed from a fall or launch, set the energy lost equal to the kinetic energy gained, m g h = one half m v squared, then solve for v with v = sqrt(2 g h).
5. For power, divide the energy transferred by time, or use P = m g v when the body moves at a steady speed.
6. Substitute numbers carefully, keep units consistent (kg, m, s), and state the final answer with its unit (J, W or m s-1).

Where the marks are

- A mark for finding the correct change in height, not the total height above a reference level.
- A mark for stating the correct energy relationship (delta Ep = m g delta h, or energy conservation) before substituting.
- A mark for a correct substitution and calculation that leads to the answer.
- A mark for the final answer with the correct unit, which is often given separately.
- A mark for the principle in derivation questions, such as writing v = sqrt(2 g h) from energy conservation before going on to the expression.

Pitfalls

- Using absolute elevation values instead of the difference in height, which gives a wrong change in potential energy.
- Forgetting the sign of delta Ep: a gain in height means energy is stored, a fall means it is released.
- Leaving out the unit J or W, or giving a speed without m s-1.
- Using v rather than v squared, or forgetting the factor one half, when converting kinetic energy to speed.
- Mixing up work done, energy change and power, so the wrong formula is used for the question asked.

Pattern 5: Gravitational field strength

Syllabus: S26-U3-SU-11 / S27-U3-SU-11

Frequency: 5 of 10 exam years; 19 marks in all.

Marks reward: Substitution and working 7, Select the principle or relationship 4, Qualitative explanation 3, Final answer 3, Units, significant figures, direction 2

Past questions: 2016P-15(e)；2018P-13(c)；2019R-4；2023P-15(b)；2025R-6

Summary: This point examines gravitational field strength, defined as the net gravitational force per unit mass at a point, with the working relationships $g = F/m$ and $g = GM/r^2$. Papers usually test it as a calculation of field strength or orbital acceleration at a given distance, a rearrangement to find a distance or mass, or a short explanation comparing field strengths on two bodies using the formula.

Recognise: Look for a question that gives the mass of a body and a distance from it, or gives the height of a satellite above a planet or moon and asks for the acceleration due to gravity. A force on a known mass in a field, asking for the field strength, is another sign. Phrases such as 'height above the surface' mean the radius must be rebuilt as body radius plus height, and words such as 'explain' or 'compare' call for reasoning through the formula rather than just a number.

Steps

1. Identify the body producing the field, its mass M, and the distance r measured from its centre. If a height above the surface is given, add the body's radius to it.
2. Convert every quantity to SI units (kg and m). Watch for km, and for masses given as multiples of Earth's mass, which must be converted first.
3. Write the relationships: $g = F/m$, where the unit N kg^{-1} is the field strength, and $g = GM/r^2$, using G from the data booklet. Note that N kg^{-1} and m s^{-2} are equivalent when acceleration is asked for.
4. If a force on a mass is given, use $F = mg$. If a distance or mass is asked for, first rearrange so that r^2 or M is the subject, then substitute.
5. Substitute the values, keep enough significant figures through the working, and evaluate.
6. For an explanation, state the formula, name which variable changes, and say how g changes as a result (g increases with mass and decreases with distance from the centre).
7. Give the final answer with correct units and an appropriate number of significant figures, in the form the question asks for.

Where the marks are

- A mark is usually given for stating the correct formula $g = GM/r^2$ or $g = F/m$ before substituting.
- Separate marks are given for converting the distance to the distance from the centre and for converting units, so each conversion step should be shown.
- A mark is given for substituting correct values into a rearranged equation even when the final number is wrong, so the working should be visible.
- The final mark goes for a correct numerical answer with the right unit, and sometimes for giving the value in two unit forms when the question asks for both.
- Explanation answers earn marks for each relevant factor: naming the mass and the radius, comparing their values, and linking the result to a weaker or stronger field. Discussing only one factor usually earns partial credit only.

Pitfalls

- Using the surface distance instead of the distance from the centre of the body, which is the most common reason for a wrong answer.
- Thinking the field depends on the mass of the object feeling it; g depends only on the body creating the field and the distance from its centre.
- Forgetting to convert km or Earth-mass multiples, or making a power-of-ten slip, which loses the calculation marks.
- Rounding too early or giving the wrong number of significant figures, so the final answer mark is lost.
- In explanations, mentioning only mass or only radius when both matter, or saying gravity is 'less' without referring to the formula.

Pattern 6: Inclined planes: components of weight

Syllabus: S26-U3-SU-12 / S27-U3-SU-12

Frequency: 4 of 10 exam years; 8 marks in all.

Marks reward: Select the principle or relationship 4, Substitution and working 3, Final answer 1

Past questions: 2016P-18(d)；2017R-11；2021R-6(b)；2025R-9

Summary: This point examines how the weight of an object on a frictionless or simple incline is split into a component along the slope and a component into the slope. Questions usually appear as a ladder, box, rope or bar on a ramp, asking for a tension, an applied force, an angle, or an acceleration, often with a second step that uses Newton's second law or a kinematics result.

Recognise: Look for an object resting on, sliding down, or being pulled up a slope, a rope or force acting parallel to the ramp, or a question asking for the angle of an incline. Any wording about a ladder used as a ramp, or a bar sliding on rails at an angle, uses the same method. The key cue is a force that is not vertical or horizontal, so you must split it into two perpendicular directions.

Steps

1. Draw a free-body diagram: weight (mg) acting vertically downward, the normal reaction perpendicular to the surface, and any tension, applied, friction or magnetic force acting along or against the slope.
2. Choose axes tilted to the incline: one parallel to the plane (down the slope positive, or in the direction of motion) and one perpendicular to it.
3. Resolve weight into components: parallel to the plane W_parallel = mg sin(theta), and perpendicular to the plane W_perp = mg cos(theta), where theta is the angle of the slope measured from the horizontal.
4. Apply equilibrium or Newton's second law perpendicular to the plane: N = mg cos(theta) if no other perpendicular forces act.
5. Apply the second law along the plane: F_net = ma. If the object is stationary or moving at constant velocity, the net force is zero, so the applied or rope force equals mg sin(theta). If it slides freely, a = g sin(theta).
6. To find an angle, rearrange the ratio, for example sin(theta) = (component along plane) / (weight), then use inverse sine. Check that the angle is sensible and that your calculator is in degree mode.

Where the marks are

- A mark for resolving the weight into components parallel and perpendicular to the plane, shown on a diagram or stated in words.
- A mark for the correct trigonometric function: sin for the component along the slope and cos for the component perpendicular to it, with theta measured from the horizontal.
- A mark for equating forces correctly for stationary or constant-velocity motion, or for setting up F = ma along the plane.
- A mark for correct substitution into the equation with consistent units, such as mass in kg and g in m s-2.
- A mark for a final answer with correct units and sensible rounding, often one more significant figure than the data.

Pitfalls

- Using sin for the perpendicular component or cos for the parallel component, which is the most common error.
- Confusing the angle of the slope with the angle of the rope or force, or measuring the angle from the slope surface rather than from the horizontal.
- Forgetting the normal reaction or ignoring it when deciding the perpendicular balance.
- Rounding intermediate values too early, which changes the final answer, or leaving the calculator in radian mode.
- Assuming the applied force equals the full weight rather than only the component along the plane, or

treating the rope as perpendicular to the slope.

Pattern 7: Projectile motion

Syllabus: S26-U3-SU-13 / S27-U3-SU-13

Frequency: 10 of 10 exam years; 120 marks in all.

Marks reward: Substitution and working 43, Final answer 42, Select the principle or relationship 26, Graph work 6, Diagram 1, Qualitative explanation 1, Evaluate and conclude 1

Past questions: 2016P-15(a); 2016P-15(b); 2016P-15(ci); 2016P-15(cii); 2016R-3; 2017P-14(a); 2017P-14(bi); 2017P-14(bii); 2017P-14(c); 2017P-14(d); 2017P-14(e); 2018P-11(a); 2018P-11(b); 2018P-11(c); 2018P-11(d); 2019C-20(ai); 2019P-12(b); 2019P-12(c); 2019P-12(d); 2020P-14(b); 2020R-1; 2020R-10; 2020R-11(a); 2021P-15(a)…

Summary: This point examines projectile motion by treating the horizontal and vertical motions separately, using the constant-acceleration equations on the vertical component and constant velocity on the horizontal component. It usually appears as a multi-part question about an object launched or thrown near the ground, asking for time of flight, horizontal range, impact or resultant velocity, and sometimes a velocity-time sketch or kinetic energy.

Recognise: Any question about an object launched at an angle, fired horizontally, kicked, thrown, or dropped from a height, where the wording asks for time in the air, distance travelled, height reached, speed or angle on landing, or a graph of its motion. Look for a given launch speed and angle, or a height above ground, and for the phrase 'ignore air resistance', which tells you gravity is the only force acting after launch.

Steps

1. Set up axes and a sign convention: choose up (or down) as positive and write down, separately for horizontal and vertical, the known initial velocity, displacement, acceleration and the quantity to find.
2. Resolve the launch velocity into components: horizontal = v cos θ and vertical = v sin θ, with θ measured from the horizontal.
3. Horizontal acceleration is zero, so the horizontal velocity stays constant and horizontal displacement is s = vH t.
4. Vertical acceleration is a = -g (about 9.8 m s-2 downward). Use v = u + at, s = ut + ½at², or v² = u² + 2as on the vertical component only. To find time in the air, set the vertical displacement equal to the change in height, which may give a quadratic in t; choose the physically sensible root.
5. Combine results: range = vH × t, impact vertical speed = |u + at| at the landing time, and resultant speed = √(vH² + vV²) with direction tan-1(vV / vH) below or above the horizontal.
6. Where energy or a graph is asked, use Ek = ½ m v² with the resultant speed at that point, and sketch the vertical velocity as a straight line decreasing at constant rate while the horizontal velocity is a flat line.

Where the marks are

- Method marks for correctly resolving the initial velocity into horizontal and vertical components, usually shown by stating or using the cosine and sine terms.
- A mark for choosing the correct vertical equation and writing the known values with correct signs before substituting.
- A mark for recognising that horizontal velocity is constant and using s = vt for horizontal distance.
- Marks for combining components (Pythagoras and inverse tangent) when the question asks for a resultant speed or angle, with the angle stated relative to the horizontal.
- Answer marks for the final value with correct units and the required precision, often awarded only if the earlier method is correct, so show intermediate lines.

Pitfalls

- Using the full launch speed in the vertical or horizontal equation instead of the component, or swapping sine and cosine when the angle is measured from the ground.
- Using the total speed to find the time of flight, or applying the constant-velocity equation vertically

where acceleration is present.

- Keeping inconsistent sign conventions, for example taking g as positive while launch is upward, which flips the sign of the vertical velocity at landing.
- Stopping at the quadratic root or taking the negative time root when the object is launched from a height, rather than choosing the physical solution.
- Giving the impact angle as measured from the vertical, or reporting only one component when a resultant velocity and direction are asked for.

Pattern 8: Orbits, satellites and Kepler's third law

Syllabus: S26-U3-SU-14 / S27-U3-SU-14

Frequency: 7 of 10 exam years; 76 marks in all.

Marks reward: Substitution and working 26, Final answer 19, Select the principle or relationship 15, Qualitative explanation 7, Evaluate and conclude 3, Graph work 3, Units, significant figures, direction 2, Recall 1

Past questions: 2016C-21(c); 2016P-15(d); 2016R-4(a); 2017P-17(a); 2017P-19(a); 2017P-19(c); 2017P-19(d); 2018C-18(a); 2018C-18(b); 2018P-13(a); 2018P-13(b); 2018P-13(d); 2020P-17(a); 2020P-17(bi); 2020P-17(bii); 2022C-19(ai); 2022C-19(aii); 2022C-19(b); 2022C-19(d); 2023P-15(a); 2023P-15(ci); 2023P-15(cii); 2025R-11

Summary: This point covers planets, moons and satellites moving in orbits modelled as uniform circular motion. Gravity is set equal to the required centripetal force, which leads to Kepler's third law (T squared proportional to r cubed, with T squared over r cubed equal to 4 pi squared over GM). In the papers it appears mainly as multi-step calculation questions (orbital speed, period, radius, altitude, or the mass of the central body), often with a short derivation or explanation part and occasionally a graph of T squared against r cubed.

Recognise: Any question about a satellite, moon, planet or geostationary orbit. Look for words such as orbital radius, period, orbital speed, altitude above the surface, barycentre, or Kepler. A request to derive or show a relationship between period and radius, or a graph of period against radius, also signals this point.

Steps

1. Find the orbital radius r, measured from the centre of the central body. If the question gives altitude above the surface, add the body's radius to it.
2. Write the force balance: gravitational force equals centripetal force, so GMm/r^2 = mv^2/r, giving v = sqrt(GM/r).
3. For period, substitute v = 2 pi r / T to obtain Kepler's third law: T^2 = (4 pi^2 / GM) r^3.
4. Convert all quantities to SI units before substituting: km to m, and hours, minutes or days to seconds.
5. Rearrange algebraically for the unknown (r, T, M or G) first, then substitute numbers once.
6. For a graph of T^2 against r^3, the gradient equals 4 pi^2 / GM, so the mass or G can be found from the gradient of the best-fit line (not from individual data points).
7. Check the size and units of the answer, and state the final result with units.

Where the marks are

- Writing the force equality (gravity supplies the centripetal force) before any algebra earns its own mark, even if the final number is later wrong.
- Converting altitude to orbital radius and time to seconds are often marked as separate steps, so show both explicitly.
- Rearranging to make the required quantity the subject, and writing the relationship in the data-sheet form, are marked steps.
- Graph questions reward using the gradient of a line and linking the gradient to 4 pi^2 / GM, together with a clear statement that the central mass is constant.
- Explanation parts reward a causal link: because GM is constant, orbital speed falls as orbital radius increases, and the reason for this should be stated, not just the trend.

Pitfalls

- Using the surface radius of the planet as the orbital radius instead of the distance from the centre of the body.
- Forgetting to convert the period from hours or days to seconds, or the altitude from kilometres to metres.
- Using individual data points instead of the gradient of the line on a graph question.
- Writing a correct final number without the force equation, which loses the principle marks.
- Taking the wrong power of r when rearranging Kepler's law (for example, confusing r cubed with r squared), or dropping the 4 pi^2 factor when isolating G or M.

Pattern 9: Artificial satellites: orbits and uses

Syllabus: S26-U3-SHE-01 / S27-U3-SHE-01

Frequency: 0 of 10 exam years; 0 marks in all.

Summary: This point asks you to match an artificial satellite's orbit (classified by altitude as low, medium or high Earth orbit, and by inclination as equatorial, polar or sun-synchronous) to its use, and to justify the choice. The input file has no past-paper awards or questions for this point (parts_total 0), so this note is written from the physics alone.

Recognise: A question names a use (communication or broadcast, navigation, weather or remote-sensing imaging, mapping the whole Earth) and asks which orbit suits it or why. Look for clues: the satellite must stay above one spot on the ground (geostationary), must cover the whole globe (polar), must pass over each place at the same local time (sun-synchronous), or needs fine detail (low altitude).

Steps

1. Identify what the orbit must do: fixed position above one spot, global coverage, repeat passes at the same local solar time, or close-range imaging.
2. Choose the altitude band. Higher orbits have a longer period and cover a larger area per satellite; low orbits give finer resolution and need less energy to reach.
3. For a circular orbit, gravity supplies the centripetal force: GMm/r^2 = mv^2/r, so v = sqrt(GM/r) and T = 2pir/v, which gives T^2 proportional to r^3.
4. Choose the inclination for ground coverage. Equatorial orbits (inclination 0) stay over the equator; polar orbits (inclination about 90 degrees) pass over both poles, and the Earth turns beneath them so the whole surface is scanned.
5. For a geostationary satellite the period must equal Earth's rotation period (24 h). It orbits in the equatorial plane at the radius where T = 24 h, moving east with the Earth, so it appears fixed in the sky.
6. For a sun-synchronous satellite use a near-polar, retrograde orbit (inclination a little more than 90 degrees). Its orbital plane precesses about the same rate the Earth moves around the Sun, so it crosses each latitude at the same local solar time.
7. Write the answer as: orbit type, altitude and inclination, then the physical feature that links it to the use.

Where the marks are

- A mark for naming an orbit type (altitude band and/or inclination) that matches the stated use.
- A mark for linking an orbit feature to the use, for example a fixed position requires a period equal to Earth's rotation period.
- A mark for correctly applying GMm/r^2 = mv^2/r or T^2 proportional to r^3 when a speed or period is asked for.
- A mark for explaining the trade-off: lower altitude gives finer detail, higher altitude gives wider coverage per satellite.
- A mark for a full justification sentence rather than a bare name of the orbit.

Pitfalls

- Confusing geostationary with polar. Geostationary orbits are equatorial with a 24 h period; polar orbits sweep over the ground.
- Calling a satellite stationary in space. It is always moving; it only looks fixed when its period matches Earth's rotation.
- Saying no force acts on the satellite. Gravity is the force providing the centripetal acceleration.
- Describing a sun-synchronous orbit without the retrograde (greater than 90 degree) inclination, or without the constant local solar time idea.
- Giving an altitude or orbit name with no link to the period, speed or coverage it implies.